VARAD DORLE

Full-Stack Architect & AI Engineer

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Versatile Full-Stack Architect, AI Engineer, and hardware enthusiast specializing in high-performance Android native applications, low-latency cross-device systems, transformer language modeling, and custom hardware integration. Proven capability in shipping complete projects from mechanical CAD and PCB design to scalable distributed web services.

CORE TECHNICAL COMPETENCIES

Mobile & Systems	Kotlin, Jetpack Compose, Android SDK & NDK, WebSockets, ZMK Firmware, Wear OS, ESP32
AI & Backend	Python, PyTorch, Transformers, Node.js, TypeScript, FastAPI, PostgreSQL, Supabase, Docker, Linux, Git
Hardware & Tools	KiCad (PCB Layout), 3D Printing, Parametric CAD, OBS API Automation, ADB, Linux Tooling

FEATURED PROJECTS

deck-8 — Open-Source Hardware Macro Pad

Hardware & Firmware

Custom 8-key mechanical macro pad with dual rotary encoders, 0.91" OLED display, Bluetooth, and battery management.

- Engineered custom multi-layer PCB in KiCad and modeled injection-grade enclosure.
- Implemented ZMK firmware supporting multi-layer keymaps, rotary navigation, and cross-device Bluetooth profiles.

StreamLink Deck — Android-to-PC Control Surface

Mobile & Distributed

High-performance native Android control deck communicating with a Node.js desktop host over low-latency WebSockets and ADB.

- Built responsive, modern UI with Jetpack Compose featuring real-time bi-directional telemetry.
- Integrated OBS WebSocket control, system macro triggers (nut-js), and automated device discovery.

CortexCAD & Compute — AI CAD & Engineering Engine

Computational Engineering

Parametric CAD copilot and volumetric computation engine utilizing AI agents and geometric processing libraries.

- Implemented interactive feature tree generation from natural language specifications and scans.
- Prototyped high-density computational geometry workflows with CT scan data and PicoGK.

Levo AI — From-Scratch Transformer Language Model

Deep Learning

Complete GPT-style autoregressive decoder model built and trained entirely from scratch in PyTorch.

- Architected multi-head causal self-attention, positional embeddings, and custom BPE tokenizer.
- Configured end-to-end local training pipeline, checkpointing, and live interactive CLI chat inference.

WristCursor — Wear OS Air Mouse

Wearable Tech

Wear OS smartwatch application transforming arm gestures into real-time PC mouse movement.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science Graduating 2026

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Computing.